SDG 15.2.3

SDG 15 Life on Land

Maintain and extend current ecosystems' biodiversity

Mariano Marcos State University · THE Sustainability Impact Ratings 2027

The question *Work directly to maintain and extend existing ecosystems and their biodiversity, of both plants and animals, especially ecosystems under threat.*

1 · Introduction

MMSU's commitment is to take an active role in protecting and promoting sustainable use of terrestrial ecosystems, responsible land use, and biodiversity conservation. Three ecosystems, one method: raise the stock locally, plant it with partners and protect what survives: forest, food plants and mangrove.

2 · Body

2.1 · Enriching MMSU's 1,068-Hectare Forest Reserves with Native Species



Bamboo planting site in the University Forest Reserve, Brgy. Mabaleng, 2 July 2025. (Photo: MMSU STRATCOM)

MMSU manages Forest Reserve Areas in Barangays Lubbot and Payao, City of Batac, covering a total of 1,068.13 hectares, where the university's forestry and agroforestry research is set up. The university maintains and extends these ecosystems with planting stock raised in its own nurseries, the MMSU Central Nursery and the MMSU Bamboo Nursery.

On July 2, 2025, 230 CAFSD graduating students enriched the reserves with 120 seedlings of the native dipterocarps guijo and yakal in Payao and 130 propagules of kawayan tinik, machiku and giant bamboo in Mabaleng. The activity was led by CAFSD OIC-Dean Charlie Batin with the MMSU Bamboo Research and Development Center.

Resources: 1, 2

2.2 · Conserving Ilocos Indigenous Food Plants In Situ, Ex Situ and Circa Situ



Partners receive indigenous food plant planting materials for the Community Foodscapes and Seedbanks, World Food Day 2025.

Through its Indigenous Food Plants (IFP) Program, MMSU conserves the wild and neglected food plants of Ilocos Norte, 46 of which were documented by the program in seven towns of the province. The program follows a triad of conservation approaches: an IFP Botanical Garden for ex situ conservation, a himbabao (alukon) genebank at the MMSU Forest Reserve, circa situ cultivation of the wild yams kamangeg and dolian, and the integration of IFPs into community foodscapes.

In October 2025, in celebration of World Food Day, the MMSU Extension and Research Directorates established the second batch of Community Foodscapes and Seedbanks for Indigenous Food Plants with local government units and schools, at Eladio V. Barangan Memorial Elementary School in San Nicolas, Ilocos Norte. The foodscape sites serve as seedbanks and learning gardens, and partner communities receive planting materials and technical assistance to cultivate kapaskapas, alokon, pako, dalayap, wild singkamas and roselle. The new partners join existing foodscape sites in Sarrat and Batac.

The IFP Program was adopted by the Regional Development Council of Region 1 as its first research and development initiative, and a study on the program by Dr. Menisa A. Antonio and co-authors won Best Paper in the Sustainable and Smart Agri-Aquaculture and Climate Resilience category of the SUC-ACAP 2025 Biennial Conference, held August 27–29, 2025, in Tagbilaran City, Bohol.

Resources: 3, 4, 5

2.3 · Nature-based Solutions for Mangrove Ecosystems in Badoc and Currimao



Mangrove site, from the gallery of the October 2025 seminar-workshop of the NbS mangrove project. (Photo: MMSU)



Dr. Arlene L. Gonzales at the MMSU–DENR Region 1 dialogue on mangrove conservation, ILAARRDEC.

Throughout 2025, MMSU implemented the DOST-PCAARRD-funded project “Nature-based Solutions (NbS) for Sustainable Mangrove Management and Resilient Coastal Communities Through Innovative Green Engineering Approach,” led by Dr. Arlene L. Gonzales, in Badoc and Currimao, Ilocos Norte, with DENR-PENRO Ilocos Norte, the Provincial Government of Ilocos Norte and the LGUs of Badoc and Currimao. The project uses green engineering and locally available, ecologically friendly materials, including bamboo-engineered structures that protect young mangroves and improve their survival.

In 2025, MMSU and DENR Region 1 met at ILAARRDEC to advance the work, with DENR pledging technical support and active participation in restoration, monitoring and evaluation, and the project trained representatives of Currimao, Pasuquin, Badoc and Burgos in mangrove propagation, management and conservation on October 27–28, 2025.

Resources: 5, 6

2.4 · MMSU, Rotary Club and Badoc LGU Restore a Typhoon-Damaged Mangrove Stand



MMSU, Rotary Club of Laoag Sunshine and LGU-Badoc plant mangrove propagules in Brgy. Pagsanahan, Badoc, 6 December 2025. (Photo: MMSU)



Participants of the mangrove enhancement and growing activity in Badoc, 6 December 2025. (Photo: MMSU)

On December 6, 2025, Mariano Marcos State University, the Rotary Club of Laoag Sunshine and the Local Government Unit of Badoc planted 280 mangrove propagules in Barangay Pagsanahan, Badoc, Ilocos Norte, to restore a mangrove stand extensively damaged by past typhoons and to ensure the continuity of ecological succession at the site.

The activity was led by MMSU President Virgilio Julius P. Manzano Jr., with the LGU of Badoc headed by Hon. Virgilio M. Calajate and the Rotary Club of Laoag Sunshine under Atty. Zsa Zsa Raval. MMSU's Vice President for Resource Generation and its Directors for Extension, Curriculum Development and Innovative Learning, and Student Affairs and Development also took part.

Resources: 7

2.5 · A New Bambusetum in Barangay Aglipay, Pinili



Participants of the bamboo propagation, nursery and plantation management training, Brgy. Aglipay, Pinili, 2025. (Photo: MMSU)

During a two-day bamboo production training in 2025, the MMSU Extension Directorate and CAFSD, with Barangay Aglipay, Pinili, and the support of ILAARRDEC, worked with 46 farmers, local leaders and community members to establish a new bambusetum, a collection of bamboo species, in Barangay Aglipay. The training covered bamboo propagation, nursery establishment and plantation management.

Resources: 8

2.6 · Indigenous Food Plants Propagated at the MMSU Botanical Garden Are Shared with Schools and Communities

Ongoing programme · Philippine News Agency, 6 June 2026



Capture: Philippine News Agency post of 6 June 2026 on indigenous food plants propagated at the MMSU NRCMC botanical garden (photos: Leilanie Adriano, PNA).



Partners receive indigenous food plant planting materials for the Community Foodscapes and Seedbanks, World Food Day 2025.

On June 6, 2026, the Philippine News Agency reported that indigenous food plants such as sugod-sugod, alokon, buga and karot are being propagated at the botanical garden of the MMSU Natural Resources Conservation and Management Center in Batac City, Ilocos Norte. The planting materials are shared with schools and community stakeholders for the development of food gardens, to boost food security in the province and provide farmers with an additional source of income.

The botanical garden is part of the ex situ arm of MMSU's Indigenous Food Plants Program, which in 2025 extended its community foodscapes and seedbanks to new partner barangays.

Resources: 9

2.7 · Mangrove Nature-based Solutions Project Runs from 2024 to 2026

Ongoing programme · 1 May 2024 to 30 April 2026



Project Leader Dr. Arlene L. Gonzales presents the project at its launch, MMSU Review Center, 27 June 2024. (Image credit: TTPD, DOST-PCAARRD)



The MMSU project team with DOST-PCAARRD, DENR-PENRO Ilocos Norte, the Provincial Government of Ilocos Norte and the LGUs of Badoc and Currimao at the launch. (Image credit: TTPD, DOST-PCAARRD)

The DOST-PCAARRD-funded project "Nature-based Solutions (NbS) for Sustainable Mangrove Management and Resilient Coastal Communities Through Innovative Green Engineering Approach," led by Dr. Arlene L. Gonzales of MMSU, was launched at the MMSU Review Center on June 27, 2024, and runs from May 1, 2024 to April 30, 2026 in Badoc and Currimao, Ilocos Norte. DENR-PENRO Ilocos Norte, the Provincial Government of Ilocos Norte and the local government units of Badoc and Currimao support the project.

The project addresses the poor hydrological conditions that keep young mangrove propagules from surviving, using technology-assisted plantations and locally available, ecologically friendly materials, including bamboo-engineered structures that protect young mangroves and improve their survival.

The 2025 activities of the project, including the dialogue with DENR Region 1 and the October 2025 mangrove propagation seminar-workshop, are described under Indicators 15.2.3 and 15.2.4.

Resources: 10

2.8 · Bambusetums Create Living Collections for Bamboo Conservation and Learning

Ongoing programme · September 2025 and September 2026



Launch of the bambusetum at the MMSU Dingras Campus, 30 September 2026. (Photo: MMSU STRATCOM)

In celebration of Bamboo Month, the MMSU Dingras Campus, through CAFSD, launched its bambusetum on September 30, 2026, to advance bamboo research, education and sustainable use. Students, faculty, field workers and stakeholders joined the planting, where For. Charlie Maine Cacactin demonstrated the proper planting of various bamboo species, and members of the Association of Young Agriculturists, Green Triangle Organization, Health and Wellness Club and Junior Foresters' Association supported the activity. The bambusetum is the second in the university, after the one at the Main Campus.

The bambusetum will serve as a living collection of bamboo species and a learning site for teaching, research, technology development and community extension, showcasing bamboo's ecological and socioeconomic benefits for food security, livelihood, environmental conservation and climate resilience. It follows the bambusetum that training participants established in Barangay Aglipay, Pinili, in 2025. CAFSD Dean Charlie Batin said the work responds to environmental problems such as climate change.

Resources: 8, 11

2.9 · Biodiversity Conservation Learning Center and Leisure Park Is Enriched with 28 Native Tree Species

Ongoing programme · since 2023

On January 13, 2023, MMSU planted 120 more seedlings of 28 native tree species and two bamboo species at the Biodiversity Conservation Learning Center and Leisure Park on the Main Campus, led by then President Shirley C. Agrupis with administrators, faculty, staff and students. The park is used as a field laboratory for classes and was identified for development into a health and wellness tourism hub, a plan endorsed by the provincial government and aligned with the Health and Wellness Complex of the College of Medicine.

The planting continues a tradition that began in 2017 with the Green Wall project on the Batac campus and extends to the Nagbacsan site and the Currimao, Dingras and Laoag campuses, as the university's contribution to the provincial "Green Wall of Ilocos Norte" and the National Greening Program of the Department of Environment and Natural Resources.

Resources: 12

3 · Conclusion

Forest reserves, indigenous food plants, mangroves and bamboo: MMSU raises planting stock in its own nurseries, plants with partners, and keeps collections and seedbanks that schools and communities can draw on.

4 · Resources

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